

# Proximal-to-Distal Energy Transfer in the Golf Swing: Where It Occurs, How It Differs with Skill, and When It Should Happen

A literature synthesis and modeling perspective grounded in the UpstreamDrift  
ZTCF/ZVCF counterfactual framework

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## Abstract

Proximal-to-distal (P→D) sequencing — the ordered peaking of pelvis, thorax, arm, and club rotational speeds — is the most widely reported organizing principle of the golf downswing. It is usually interpreted as evidence that mechanical energy flows outward along the kinetic chain, and it is often translated into the coaching prescription that energy should be passed distally as early and as completely as possible. This article reviews the mechanics and the empirical literature behind that interpretation and asks a sharper question: *when* during the downswing should the P→D transfer occur? Classical double-pendulum analyses, delayed-release simulation studies, and optimal-control results consistently favor a strategy in which energy is first accumulated in the proximal segments (and in the arms-plus-club system rotating as a quasi-rigid body), with the transfer to the club deliberately *restrained* early in the downswing and *accelerated* late. Empirical skill-level comparisons show that expert golfers differ from amateurs more in transition timing, segment-speed magnitudes, and transfer efficiency than in the mere presence of a P→D order, and methodological work cautions that peak-ordering alone is a weak proxy for energy flow. We connect this literature to the zero-torque counterfactual (ZTCF) and zero-velocity counterfactual (ZVCF) decompositions implemented in the UpstreamDrift platform, which separate, at every instant of a measured swing, the acceleration attributable to accumulated motion (the drift term) from the acceleration attributable to instantaneous muscular effort (the control term). We formalize the “retain-early / release-late” hypothesis in counterfactual terms, inventory the models and energy-transfer computations currently available in the repository, identify the measurement gaps (notably segmental energy-flow accounting), and specify falsifiable simulation experiments — torque-timing sweeps, counterfactual energy budgets, and re-simulation parity studies — that can decide the question quantitatively. No new experimental data are reported; every quantitative claim herein is sourced to the cited literature or to inspectable code in the repository.

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## **i** Provenance and scope

This article is the deliverable of UpstreamDrift epic [#8373](#) (sub-issues [#8374](#)–[#8379](#)). It is a *review and modeling-perspective* paper: it reports no new experimental data. Sections that propose experiments (Section 9) describe planned work tracked in issue [#8377](#); their results are deliberately absent here. All references were verified against publication records at the time of writing; repository links point to the **main** branch of [D-sorganization/UpstreamDrift](#).

# 1 Introduction

Clubhead speed at impact is the dominant controllable determinant of drive distance, and the golf downswing is one of the most-studied examples of a *sequenced* multi-segment rotation in sport biomechanics ([Hume, Keogh, and Reid 2005](#); [McPhee 2022](#)). Across measurement systems and populations, the downswing of a skilled golfer shows a characteristic order: the pelvis reaches its peak angular speed first, then the thorax, then the lead arm, and finally the club, with each successive segment reaching a higher peak speed than the one before it ([Cheetham et al. 2008](#); [Tinmark et al. 2010](#)). This pattern — the *kinematic sequence*, an instance of the general proximal-to-distal (P→D) sequencing observed in throwing and striking skills ([Putnam 1993](#)) — is routinely interpreted as the visible signature of mechanical energy being generated in the large proximal segments and transferred outward to the small, fast distal ones.

That interpretation immediately raises a question that the descriptive literature does not settle:

**Should a golfer attempt to transfer energy from proximal to distal segments *early* in the downswing, or should energy be *retained* in the proximal segments early and the transfer deliberately *accelerated* late?**

The question is not academic. The two strategies imply opposite coaching cues (e.g., “fire the club from the top” versus “hold the lag and let it release”), opposite torque programs at the wrists and torso, and different predictions for what counterfactual decompositions of a measured swing should show. It also connects to a long-standing puzzle in the modeling literature: simple double-pendulum analyses have argued since the 1960s that the highest clubhead speeds arise when the distal joint is *not* driven early — indeed when the release of the club is passively or actively delayed (Cochran and Stobbs 1968; T. Jorgensen 1970; Milburn 1982; Pickering and Vickers 1999) — while the coaching translation of the kinematic sequence often emphasizes initiating distal motion promptly after transition.

The UpstreamDrift project approaches this question with *counterfactual dynamics*. Two decompositions are implemented across its physics engines:

- the **zero-torque counterfactual (ZTCF)** — the acceleration the system would exhibit at the current instant if all applied joint torques were removed, isolating what accumulated motion, gravity, and inter-segment coupling produce on their own; and
- the **zero-velocity counterfactual (ZVCF)** — the acceleration the applied torques would produce at the current pose if the system were momentarily at rest, isolating instantaneous muscular effort from motion-dependent effects.

Preliminary ZTCF/ZVCF analyses conducted with the repository’s MATLAB pipeline (see Section 6) motivated the hypothesis examined in this article: that *restraining* the P→D handoff early in the downswing — keeping kinetic energy in the proximal segments and in the arms-plus-club system rotating as a quasi-rigid unit — and accelerating the handoff late, may be the more effective strategy. This article evaluates that hypothesis against the published literature and specifies how it can be tested quantitatively with the tools already present in the repository.

**Contributions.** The article makes five contributions:

1. a synthesis of the mechanics of sequenced swings, from the summation-of-speed principle to segment-interaction dynamics and parametric energy transfer (Section 2);
2. a methodological account of what “energy transfer” means operationally — joint power, joint-force power, segmental energy accounting, and induced accelerations (Section 3);
3. a critical review of the empirical evidence on where P→D transfer occurs and how it differs between skilled and less-skilled players (Section 4);
4. a formalization of the early-versus-late transfer question, including the ZTCF/ZVCF counterfactual framework and the specific predictions the “retain-early / release-late” hypothesis makes (Section 5, Section 6); and
5. an inventory of the models and computations available in UpstreamDrift today, an honest gap analysis, and a set of falsifiable simulation experiments that would decide the question (Section 7, Section 9).

## 2 Mechanics of sequenced swings

### 2.1 The summation-of-speed principle and its limits

The oldest formalization of P→D sequencing is the *summation of speed* principle: in skills demanding high end-point speed, each segment should begin its motion at the moment of peak speed of the segment proximal to it, so that distal speed is “built on top of” proximal speed (Bunn 1972). Milburn applied the principle directly to golf with a double-pendulum analysis of the downswing and found the characteristic pattern of an initially dominant proximal (arm) rotation followed by rapid distal (club) rotation, noting that “*an initial delay in the uncocking of the wrist*” allows the arm to reach greater acceleration and the club’s acceleration to be summed with the proximal segment’s maximum (Milburn 1982). It is worth pausing on the structure of that finding: the summation principle, taken naively, prescribes *sequential initiation*; Milburn’s analysis showed the benefit comes specifically from *delaying* the distal contribution, not merely ordering it.

Putnam’s segment-interaction analyses put the principle on a dynamical footing and simultaneously exposed its limits (Putnam 1991, 1993). In a linked chain, each segment’s angular acceleration depends not only on the joint torques but on *motion-dependent interaction torques* — centripetal and Coriolis terms proportional to the squares and products of segment angular velocities. Putnam showed that the P→D pattern in striking and throwing arises to a substantial degree from these interaction torques: proximal-segment deceleration and distal-segment acceleration late in the motion are coupled through the joint force, and they occur even without a distal joint torque. Two implications matter for golf. First, the kinematic order does not by itself reveal *which* torques produced it: an identical peak sequence can be produced by different mixes of muscular and interaction torques. Second, the deceleration of proximal segments observed in experts need not be an active “braking” strategy — it can be, at least in part, the dynamical *consequence* of the distal segment accelerating and back-reacting on the chain. Marshall and Elliott further cautioned that planar peak-ordering analyses miss long-axis rotations, which peak very late in racquet skills and contribute substantially to end-point speed (Marshall and Elliott 2000); the golf analogue is forearm rotation and wrist action in the final phase of the downswing.

## 2.2 Double-pendulum mechanics of the golf downswing

The two-lever, one-hinge model of the downswing — an “arm” link rotating about a fixed or moving hub, connected by a wrist hinge to a “club” link — has been the workhorse of golf mechanics since Cochran and Stobbs’ *Search for the Perfect Swing* (Cochran and Stobbs 1968) and Jorgensen’s dynamical analysis fitted to stroboscopic photographs of a professional swing (T. Jorgensen 1970; T. P. Jorgensen 1999). Its enduring value for the present question is that it makes the *timing* of the P→D handoff an explicit, controllable variable (the “release”: the point in the downswing at which the wrist angle is allowed, or driven, to open) and exposes the mechanism by which delay pays.

Three results from this tradition anchor the analysis:

**Delayed release increases clubhead speed.** Pickering and Vickers, sweeping release angles in a driven double-pendulum model, found clubhead velocity at impact increases as release is delayed — the classical “late hit” (Pickering and Vickers 1999). Jorgensen’s fitted model similarly reproduced the expert swing only with a substantially delayed opening of the wrist angle (T. Jorgensen 1970; T. P. Jorgensen 1999). In these models an *early* release (early P→D handoff) is strictly inferior: club kinetic energy purchased early is paid for with arm speed that never develops.

**Late wrist torque helps; early wrist torque is the wrong tool.** Sprigings and Neal, using a torque-driven three-segment model with physiologically bounded actuators, found that an active wrist torque applied *during the latter stages of the downswing* increased clubhead speed by approximately 4% (Sprigings and Neal 2000); Sprigings and MacKenzie confirmed the delayed-release benefit in a follow-up simulation focused specifically on the timing of release (Sprigings and MacKenzie 2002). The point is not that distal effort is useless, but that its productive window is late.

**Energy transfer to the club can be largely passive — and is governed by the changing geometry.** White analyzed an *undriven* double pendulum and showed that the transfer of kinetic energy from arms to club late in the downswing is a parametric-transfer phenomenon: as the club swings out, the effective moment of inertia of the system changes, and conservation of energy and angular momentum move kinetic energy distally without any wrist torque at all (White 2006). The efficiency of this transfer is controlled principally by the wrist-cock angle maintained before release (White 2006). Miura’s analysis of the “inward pull” complements this: an expert’s hands moving toward the body near impact — equivalent to shrinking the hub radius when centrifugal loading is fully developed — performs work against the centrifugal force that appears as additional clubhead kinetic energy (Miura 2001). Nesbit and McGinnis’ hub-path analyses generalize the same idea: the non-circular hand path of skilled golfers, including its late-downswing tightening, is a mechanism for shaping kinetic transfer from golfer to club (Nesbit and McGinnis 2009; Nesbit and McGinnis 2014).

Together these results describe a machine whose highest-output operating mode is: *load the proximal rotation first, hold the distal joint folded (retaining the system’s energy in a compact, fast-rotating configuration), then let geometry and — in the last phase — modest distal torque move the energy outward rapidly.* In

the vocabulary of this article, the double-pendulum literature is already an argument for **retain-early / release-late**.

## 2.3 Optimal control and torque sequencing

If the timing question is posed as an optimization — what torque program maximizes clubhead speed (or carry) subject to actuator limits? — the answer has been studied repeatedly. Lampsa’s early optimal-control study of a driven double pendulum computed torque histories maximizing drive distance, and notably found the *optimal* torques differed in structure from those measured (via inverse dynamics) in his own swing (Lampsa 1975) — early evidence that human swings need not be optimal and that the optimum is a meaningful target rather than a redescription of expert behavior. Sharp, fitting shoulder-arm-club models to expert swing data and then optimizing torque programs, established a pattern for the best use of the available driving torques across the downswing phases, with the distal contribution structured late rather than distributed evenly (Sharp 2009). Three-dimensional forward-dynamics models extended the approach with anatomically meaningful actuators and flexible shafts: MacKenzie and Sprigings’ 3D model (MacKenzie and Sprigings 2009), the carry-optimized model of Balzerson, Banerjee and McPhee (Balzerson, Banerjee, and McPhee 2016), and the six-degree-of-freedom dynamic optimization of McNally and McPhee (McNally and McPhee 2018) all generate expert-like clubhead speeds with torque programs whose distal elements act late in the downswing. McPhee’s review surveys this modeling lineage (McPhee 2022).

A caution applies in the other direction as well: the optimal-control literature optimizes *within a model*, and the sensitivity of the optimal timing to model structure (hub mobility, shaft flexibility, actuator rate limits) is exactly why a multi-engine platform with cross-engine validation (see Section 7) is the right instrument for revisiting the question.

## 3 What “energy transfer” means operationally

Claims about energy transfer require an accounting framework. Four levels of analysis appear in the literature, and they are not interchangeable.

### 3.1 Joint power and work

For a joint spanning segments  $i$  and  $i+1$  with net joint torque  $\tau$  and relative angular velocity  $\omega$ , the *joint (moment) power* is  $P_m = \tau \omega$ ; its time integral over a phase is the mechanical work done by the net moment. Positive work indicates generation (or concentric action), negative work absorption. This is the quantity most golf studies report, and it is the basis of the work-power budget of Nesbit and Serrano (Section 4.1) (Nesbit and Serrano 2005).

### 3.2 Joint-force power and segmental energy accounting

Joint moments are not the only conduit of mechanical energy: the joint reaction force  $\mathbf{F}$  acting at a joint moving with velocity  $\mathbf{v}$  transmits *joint-force power*  $P_f = \mathbf{F} \cdot \mathbf{v}$  from one segment to the next. Robertson and Winter’s classic formulation balances, for each segment, the rate of change of segment mechanical energy against the sum of joint-force powers and moment powers at its proximal and distal ends, distinguishing energy *transferred between* segments from energy *generated or absorbed* at joints (Robertson and Winter 1980). This is the correct level of analysis for the question posed here, because the “retain-early / release-late” hypothesis is precisely a claim about the *time course of inter-segment transfer terms*, not about peak angular velocities. As Section 7.6 notes, this accounting is the principal measurement gap in the UpstreamDrift Python stack today (the MATLAB pipeline’s per-joint linear-plus-angular work decomposition is its closest existing realization).

### 3.3 Induced accelerations and dynamic coupling

In a coupled chain, any single torque induces accelerations at *every* joint — including joints it does not span — through the inverse inertia matrix (Zajac and Gordon 1989). Induced-acceleration and power-

transfer analyses in locomotion research (Zajac, Neptune, and Kautz 2002) show that intuitions of the form “muscle X accelerates joint Y” routinely fail in coupled systems. For golf this matters because a torso torque early in the downswing partly accelerates the *club* (through coupling), and a wrist torque late in the downswing back-reacts on the torso; the net energy routing is an emergent property of the whole chain. The ZTCF/ZVCF decompositions of Section 6 are the counterfactual cousins of this analysis: rather than attributing acceleration to individual torques, they split the total acceleration into a motion-dependent (drift) part and a control part, instant by instant.

### 3.4 Kinematic sequence as a proxy — and its failure modes

Peak-ordering of segment angular speeds is a *proxy* for energy flow, not a measurement of it. Marsan and colleagues showed that the identified kinematic sequence in golf depends materially on which angular-velocity component (or norm, or projection) is analyzed: thirteen golfers analyzed with seven component-selection methods yielded different sequences (Marsan et al. 2019). Marshall and Elliott’s long-axis-rotation caution (Marshall and Elliott 2000) is a special case of the same problem. Any serious test of the timing hypothesis must therefore be run at the level of energy and power accounting (Section 3.2), with the kinematic sequence reported as corroborating description only.

## 4 Empirical evidence in golfers

### 4.1 The work and power budget of the swing

Nesbit’s full-body kinematic-kinetic study and the companion work-power analysis of Nesbit and Serrano remain the most complete published energy budgets of the golf swing (Nesbit 2005; Nesbit and Serrano 2005). Using subject-specific full-body models of four golfers of widely differing skill, they report that the trunk complex (lumbar and thoracic spine plus hips) contributes on the order of 68.7–72.2% of the total work of the swing, the shoulder and arm joints 24.3–28%, and the legs only 3.3–3.8% (Nesbit and Serrano 2005). Two further observations from the same study bear directly on this article’s question. First, power generation and clubhead speed decreased systematically with increasing handicap. Second, the authors’ phase analysis locates the *generation* of energy predominantly in the earlier downswing (torso-dominated) and the *conversion* into clubhead speed late (shoulder/arm and club interface) — i.e., generation and transfer are temporally separated, which is a precondition for the retain-early / release-late strategy to even be possible.

The ground is the ultimate source of the external impulse. Han and colleagues characterized golfer-ground interaction in 63 skilled golfers and found meaningful associations between ground-interaction force and moment parameters and maximum clubhead speed (Han et al. 2019). Recent work has pushed this to an explicitly energetic formulation: in thirty golfers (fifteen professionals, fifteen high-level amateurs), foot-ground interaction variables alone explained 35.5% of clubhead-speed variance, rising to 75.4% when trunk sequencing and an *impulse-based energy-transfer efficiency* measure (effectiveness of mechanical impulse transmission from proximal segments to the club) were added; mediation analysis attributed the dominant indirect pathway to transfer efficiency rather than to sequencing per se (“Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026). That last distinction — transfer *efficiency* mattering where sequence *order* alone does not — is a recurring theme of the skill-level literature reviewed next.

### 4.2 Kinematic-sequence evidence and skill-level differences

The descriptive P→D findings are robust in skilled populations. Tinmark and colleagues, studying elite golfers across full and partial shots, found a significant proximal-to-distal temporal ordering of peak speeds (pelvis → torso → hands/club) with successive amplification of peak speed at every level of effort, in both men and women (Tinmark et al. 2010). Cheetham and colleagues compared amateur recreational players with touring professionals on transition- and downswing-phase kinematic sequence parameters and found professionals showed higher peak-speed magnitudes and more consistent sequencing/timing characteristics (Cheetham et al. 2008). Zheng and colleagues’ three-dimensional comparison of professional and amateur

swings found systematic temporal and spatial differences, with amateurs deviating progressively more from professional norms as handicap increased (Zheng et al. 2008). Meister and colleagues provide elite rotational benchmarks (pelvis-thorax separation and its timing) against which amateurs can be compared (Meister et al. 2011), and Lindsay, Mantrop and Vandervoort review the broader catalogue of skill-level biomechanical differences (Lindsay, Mantrop, and Vandervoort 2008).

Three refinements prevent an overly tidy reading:

1. **Sequence order is common but not universal — even among the best.** Kwon’s laboratory and others have documented skilled golfers whose peak ordering deviates from the canonical pelvis-thorax-arm-club pattern, and recent work has examined how alternative sequence patterns among elite players relate to clubhead speed (Lee, Ehrlich, and Cain 2026). On-plane analyses of the “axle-chain” system in 66 skilled golfers found clubhead speed associations concentrated in specific angular-motion characteristics rather than in gross ordering (Madrid et al. 2020).
2. **Within-player timing differences do not straightforwardly explain shot quality.** Neal and colleagues, comparing well-timed and mistimed shots *within* highly skilled players, found no significant differences in the timing lags between peak angular speeds of contiguous segments (Neal et al. 2007) — the felt sense of “timing” apparently lives somewhere other than the inter-peak lags of a five-segment model.
3. **Method sensitivity.** As noted in Section 3, the identified sequence depends on component-selection methodology (Marsan et al. 2019), so cross-study comparisons of ordering statistics carry real uncertainty.

The empirical picture that survives these refinements: *skilled golfers generate larger proximal speeds, separate transition from release more distinctly, and transmit impulse/energy to the club more efficiently; merely exhibiting a  $P \rightarrow D$  peak order is neither unique to skill nor sufficient for it* (Cheetham et al. 2008; Zheng et al. 2008; “Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026; Lee, Ehrlich, and Cain 2026).

### 4.3 Where the transfer occurs

Synthesizing Section 4.1 and Section 4.2 with the modeling results of Section 2: in a skilled swing, the dominant energy *generation* occurs in the trunk-hip complex during the first two-thirds of the downswing (Nesbit and Serrano 2005); the dominant energy *transfer to the club* occurs in the final phase, through the wrist interface, by a mechanism that is substantially parametric/geometric (release of the wrist-cock under centrifugal loading, hub-path tightening) with a modest late active wrist contribution (White 2006; Miura 2001; Sprigings and Neal 2000; Nesbit and McGinnis 2009). Less-skilled players truncate exactly this separation: early release (“casting”) moves energy into the club before proximal speed has peaked, producing the lower distal peak speeds and lower transfer ratios observed in amateur cohorts (Cheetham et al. 2008; Zheng et al. 2008; Milburn 1982).

## 5 The timing question, stated precisely

### 5.1 Two strategies

Let  $E_p(t)$  be the kinetic energy of the proximal subsystem (trunk and arms, including the club moving with the arms as a folded unit) and  $E_c(t)$  the kinetic energy of the club, over a downswing  $t \in [t_0, t_{\text{imp}}]$ . Define the transfer functional as the inter-segment power crossing the wrist interface (joint-force power plus moment power, in the sense of Section 3.2),  $P_w(t)$ . The two strategies are:

- **S1 (transfer early):** make  $P_w(t)$  large as soon after transition as possible; the club begins acquiring its final energy early, and the golfer continues adding energy to an already-fast club.
- **S2 (retain early, release late):** keep  $P_w(t)$  small during the first part of the downswing (the wrist interface transmits centripetal force but little power; the club rides with the arms), while proximal torques maximize  $E_p$ ; then allow/drive a large  $P_w$  concentrated in the final phase.

The double-pendulum, delayed-release and optimal-control literatures reviewed in Section 2 uniformly favor S2 within their model classes (Milburn 1982; Pickering and Vickers 1999; Sprigings and Neal 2000; Sprigings and MacKenzie 2002; Sharp 2009; White 2006). The mechanistic reasons compose into a coherent account:

1. **Inertia management.** With the wrist cocked, the system’s moment of inertia about the hub is small, so proximal torque buys angular velocity cheaply; releasing early raises the inertia while proximal torque is still trying to accelerate the system (White 2006; T. P. Jorgensen 1999).
2. **Actuator physiology.** Torque-velocity limits mean the wrist’s small muscles can only add meaningful power while wrist angular velocity is modest — i.e., late torque acts over a brief window but at high force transmission; early wrist torque squanders the window and can even require *negative* (restraining) torque later (Sprigings and Neal 2000).
3. **Parametric transfer needs developed centrifugal loading.** The passive outward flow of energy at release, and the inward-pull gain, are both proportional to the centrifugal field established by proximal rotation; they are weak early and strong late (White 2006; Miura 2001).
4. **Interaction torques do the late work for free.** Putnam-style motion-dependent torques accelerate the distal segment and decelerate the proximal one automatically once release begins; the golfer’s job is to *time* this cascade, not to muscle it (Putnam 1993).

Empirically, S2-like signatures separate skill groups (later release, larger proximal speeds before release, higher transfer efficiency; Section 4.2), while the within-expert null result on inter-peak lags (Neal et al. 2007) suggests that once S2 is grossly achieved, the residual performance variance lives in finer variables (release profile shape, hub-path geometry, impulse-transfer efficiency) — exactly the variables the counterfactual framework below is designed to expose.

## 5.2 What would falsify S2?

To keep the question scientific rather than rhetorical, we state what evidence would count against the retain-early/release-late hypothesis:

- an optimal-control solution, in a validated 3D model with physiological actuator limits, whose optimal wrist-interface power is front-loaded;
- measured expert swings in which early wrist-interface power (normalized) is *positively* associated with clubhead speed across players;
- counterfactual (ZTCF/ZVCF) analyses showing that in faster swings the control term dominates club acceleration *early* rather than late.

The proposed experiments of Section 9 operationalize these tests.

# 6 Counterfactual decompositions: ZTCF and ZVCF

## 6.1 Definitions

Write the chain’s equation of motion in standard manipulator form:

$$M(q)\ddot{q} = \tau - b(q, \dot{q}), \quad b(q, \dot{q}) = C(q, \dot{q})\dot{q} + g(q) + d(\dot{q}), \quad (1)$$

with inertia matrix  $M$ , applied generalized torques  $\tau$ , Coriolis and centripetal terms  $C\dot{q}$ , gravity  $g$ , and velocity-dependent dissipation  $d$ . Two counterfactual accelerations are defined *pointwise* at each measured state  $(q, \dot{q}, \tau)$  of a swing:

$$\ddot{q}_{\text{ZTCF}} = M(q)^{-1}(-b(q, \dot{q})), \quad (2)$$

$$\ddot{q}_{\text{ZVCF}} = M(q)^{-1}(\tau - b(q, \mathbf{0})) = M(q)^{-1}(\tau - g(q)), \quad (3)$$

where the simplification in Equation 3 holds because the Coriolis term (quadratic in velocity) and viscous dissipation (linear in velocity) vanish at  $\dot{q} = \mathbf{0}$ . The ZTCF (Equation 2) answers “*what would the system do right now if every actuator switched off?*” — it is the **drift** field of the affine control system, containing gravity plus all motion-dependent interaction dynamics. The ZVCF (Equation 3) answers “*what would the current muscular effort produce from a standstill in this pose?*” — it isolates the instantaneous contribution of control from the contribution of accumulated motion. Because Equation 1 is affine in  $\tau$ , the actual acceleration superposes exactly:

$$\ddot{q} = \underbrace{M^{-1}(-b(q, \dot{q}))}_{\text{drift} = \text{ZTCF}} + \underbrace{M^{-1}\tau}_{\text{control}}, \quad (4)$$

an identity enforced to numerical tolerance by the repository’s test suite (Section 7.5).

## 6.2 Two operational variants

The repository implements two distinct operationalizations, and the difference matters for interpretation:

**Pointwise (instantaneous) counterfactuals.** The engine-agnostic Python implementation ([simulation\\_backends/ztcf\\_](#)) evaluates Equation 2 and Equation 3 at each sample of an already-measured trajectory. It is explicitly *not* a forward-integrated rollout: each sample answers a local “what if” without propagating the hypothetical forward. This is the right tool for constructing drift/control *time profiles* along a real swing.

**Re-simulation counterfactuals.** The original MATLAB/Simscape pipeline ([run\\_ztcf\\_simulation.m](#)) re-simulates the full model with a torque “killswitch” applied at time  $t$ , for a sweep of  $t$  values, tabulating the counterfactual state reached with torques off. The pipeline forms the difference table  $\text{DELTA} = \text{BASE} - \text{ZTCF}$  ([process\\_data\\_tables.m](#)), attributing to active control everything the passive system would not have done. A companion ZVCF model applies the measured torques to the same pose with velocities zeroed ([SCRIPT\\_ZVCF\\_GENERATOR.m](#)); run without gravity, the ZVCF and ZTCF contributions superpose to the BASE motion, mirroring Equation 4 at the level of simulated trajectories. The MATLAB pipeline additionally computes, for every joint and every table (BASE, ZTCF, DELTA), the linear work  $\int \mathbf{F} \cdot \mathbf{v} dt$ , angular work  $\int \tau \cdot \omega dt$ , total work and power, and the fractional contributions ZTCF/BASE and DELTA/BASE ([calculate\\_work\\_impulse.m](#), [calculate\\_total\\_work\\_power.m](#)) — precisely the Robertson-Winter-style joint-interface accounting (Section 3.2) that the timing question needs, including a kinematic-sequence plot computed *separately for the BASE, ZTCF and DELTA components* ([SCRIPT\\_324\\_PLOT\\_ZTCF\\_KinematicSequence.m](#)).

## 6.3 The counterfactual form of the timing hypothesis

The ZTCF/ZVCF language makes the S2 hypothesis crisp. Along a measured downswing, project the drift and control accelerations onto the club’s speed-producing direction and integrate the associated powers. S2 predicts, for high-performing swings:

- **Early downswing:** the *control* term should dominate energy input at proximal joints, while at the wrist interface the control contribution to club angular acceleration should be small or even negative (restraining lag against the developing centrifugal field). The ZTCF profile of the club should show that, had torques been removed, the club would have released outward — i.e., the drift field already “wants” to open the wrist, and control is *holding energy back*.
- **Late downswing:** the *drift* term (interaction dynamics: parametric transfer, centrifugal release) should dominate club acceleration, with a brief late window of positive wrist control power ([Springs and Neal 2000](#)). The DELTA (active) share of club-side work should be concentrated before release; the ZTCF (passive) share should account for most club energy gain after release ([White 2006](#)).

This is the pattern the repository’s preliminary MATLAB analyses motivated, and it is stated here as a *hypothesis to be quantified*, not a result: the figures and tables that would substantiate it are the deliverable of issue [#8377](#). Conversely, if high-performing swings show early-dominant wrist control power, S2 is falsified in exactly the terms of Section 5.

## 7 What UpstreamDrift computes today

This section inventories the repository’s relevant capabilities, so that the proposed experiments can be read against what exists. Links point to the `main` branch.

### 7.1 Models

Table 1: Golf swing models available in the repository.

| Model                                                  | Location                                             | Notes                                                                     |
|--------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------|
| Double pendulum (arms + club, clubhead tip mass)       | <a href="#">pendulum_simulator/physics.py</a>        | Lagrangian dynamics; viscous + Coulomb friction; analytic $M$ , $C$ , $g$ |
| Triple pendulum                                        | <a href="#">pendulum_simulator/physics_triple.py</a> | Three-segment planar chain                                                |
| 8-DOF closed-loop golfer (two arms + shared club)      | <a href="#">pendulum_simulator/physics_golfer.py</a> | Holonomic loop constraints, constraint-consistent dynamics; JAX variant   |
| Backend-neutral golf model (ODE / MuJoCo / MJX / Warp) | <a href="#">simulation_backends/</a>                 | One parameter set rendered to every backend; shared trace schema          |
| MuJoCo humanoid golfer                                 | <a href="#">mujoco_humanoid_golf/</a>                | Full-body model with club                                                 |
| Simscape 2D/3D golf models                             | <a href="#">Simscape_Multibody_Models/</a>           | Source of the original ZTCF/ZVCF methodology                              |
| Flexible shaft (FEM)                                   | <a href="#">physics/flexible_shaft.py</a>            | Beam model; strain energy not yet in the energy budget                    |

### 7.2 Counterfactual stack

The normative contract lives in the engine-core dynamics interface ([\\_dynamics\\_interface.py](#)): `compute_drift_acceleration`, `compute_control_acceleration`, `compute_ztcf`, `compute_zvcf`, with the superposition identity (Equation 4) stated as a contract. Engine implementations exist for MuJoCo, Drake, Pinocchio, OpenSim, MyoSuite, the analytic pendulum engines, and the Simscape bridge; the capability matrix is documented in [engine\\_capabilities.md](#). On top of these sit:

- the pointwise engine-agnostic evaluator with HDF5 persistence ([ztcf\\_zvcf.py](#));
- a MuJoCo delta-based counterfactual analyzer reporting torque-attributed and velocity-attributed acceleration effects ([counterfactuals.py](#));
- a Newton-Euler ZTCF force module with an analytic single-pendulum validation identity ([biomechanics/ztcf.py](#));
- induced-acceleration analyzers (gravity / drift / control / total components) ([induced\\_acceleration.py](#));
- an orchestration layer with capability gating and a REST/UI surface ([analysis/orchestrator.py](#));
- the MATLAB re-simulation pipeline described in Section 6.

### 7.3 Energy and power computations

- **Inter-segment power flow (MuJoCo):** [power\\_flow.py](#) computes per-DOF joint powers  $\tau\omega$ , work decomposed into drift and control shares, per-body kinetic and potential energies via center-of-mass Jacobians, per-body parent/child transfer bookkeeping, and an energy-conservation residual derived from the equations of motion.
- **Joint work/power/impulse metrics (engine-agnostic):** [power\\_work\\_metrics.py](#) (positive/negative/net work, generation-absorption splits, impulse, quasi-stiffness); [energy\\_metrics.py](#) (system energy summaries).

- **Conservation monitoring:** [energy\\_monitor.py](#) enforces drift bounds on total energy during integration (note: its potential-energy term is a documented first-order approximation, adequate for drift detection only).
- **MATLAB energetics:** the per-joint linear/angular work and power decomposition across BASE/ZTCF/DELTA tables (Section 6) — currently the richest joint-interface energy accounting in the repository.

## 7.4 Sequencing analytics

- [kinematic\\_sequence.py](#): peak detection, ordering consistency against a caller-supplied expected order, timing gaps, distal/proximal speed-gain ratios, post-peak deceleration rates. (Deliberately method-neutral; the expected order is user-defined.)
- [coordination\\_metrics.py](#): vector-coding coupling angles, continuous relative phase, pairwise lag matrices — quantitative sequencing measures less brittle than peak-ordering.
- Plot renderers for sequence traces, peak-timing Gantt charts, and lag matrices ([\\_coordination\\_sequence.py](#)).
- An approved internal research review of the kinematic-sequence metric, including literature benchmarks and uncertainty-reporting guidance ([kinematic\\_sequence.md](#)).

## 7.5 Validation evidence

The counterfactual stack is contract-tested: ZTCF/ZVCF from independent backends agree with analytic ground truth to  $10^{-9}$  ([test\\_ztcf\\_zvcf.py](#)); drift-plus-control reconstructs the measured acceleration ([test\\_counterfactual\\_experiments.py](#)); work-energy-theorem and conservation-law property tests run in CI ([tests/integration/conservation\\_laws/](#)); and a cross-engine comparison service produces divergence-annotated ZTCF/ZVCF panels between backends ([cross\\_engine\\_comparison.md](#)). The project’s design guidelines make the drift-control decomposition and ZTCF/ZVCF availability normative across engines ([project\\_design\\_guidelines.qmd](#)).

## 7.6 Gaps relevant to this article

Table 2: Known gaps (tracked in issue [#8378](#)).

| Gap                                                                                             | Consequence for the timing question                                                                                 |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Two ZTCF operationalizations (pointwise vs re-simulation) coexist; no Python re-simulation ZTCF | Analyses must state which variant they use; parity between variants is untested                                     |
| Inter-segment power transfer implemented only for MuJoCo, as joint power $\tau\omega$           | Wrist-interface <i>force</i> power $\mathbf{F} \cdot \mathbf{v}$ — central to S1/S2 — is not yet computed in Python |
| No Robertson-Winter segmental energy-flow accounting in Python                                  | The transfer functional $P_w(t)$ must currently come from the MATLAB tables                                         |
| <b>ConservationMonitor</b> potential energy is first-order approximate                          | Do not report absolute PE from it; use engine energy accessors                                                      |
| Flexible-shaft strain energy absent from the energy budget                                      | Late-downswing shaft recoil energy is invisible to current accounting                                               |
| Proximal-braking-efficiency metric is an open TODO                                              | Deceleration-side claims must rely on generic post-peak deceleration rates                                          |

## 8 Synthesis: early or late?

**What is established.** (i) Skilled golf swings exhibit P→D ordering with successive speed amplification, robustly across shot types and populations ([Tinmark et al. 2010](#); [Cheetham et al. 2008](#)). (ii) The trunk-hip complex does most of the work; the club acquires most of its energy late, through the wrist interface ([Nesbit](#)

and Serrano 2005; Nesbit 2005). (iii) In every published double-pendulum, delayed-release, and optimal-control analysis, delaying the distal handoff increases clubhead speed, and productive distal torque is late (Milburn 1982; Pickering and Vickers 1999; Sprigings and Neal 2000; Sprigings and MacKenzie 2002; Sharp 2009). (iv) The late transfer is substantially passive — parametric/geometric — once proximal rotation has established the centrifugal field (White 2006; Miura 2001). (v) Skill differences appear in magnitudes, transition timing, and transfer efficiency more than in gross order (Cheetham et al. 2008; Zheng et al. 2008; “Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026).

**What this implies for the question.** The weight of evidence supports **S2: retain energy proximally early in the downswing and accelerate the transfer late**. The phrase “retain energy proximally” deserves precision: it does not mean suppressing proximal motion — proximal segments should be accelerated as hard as the movement budget allows — it means *deferring the handoff*: keeping the wrist interface transmitting force but little power (club folded, riding with the arms) until late, so that the energy pool is large and the centrifugal field strong when the parametric transfer is allowed to run. In counterfactual terms: early control effort should appear at proximal joints and, at the wrist, partly as *restraint* of the drift field’s tendency to release; late club energy gain should be drift-dominated with a brief late window of positive wrist control power (Section 6).

**What remains genuinely open.** (a) The published S2 evidence is model-based or observational; a *direct* energetic accounting of the wrist interface across skill levels (the  $P_w(t)$  time course) has not been reported in the literature reviewed here — the closest is the impulse-transfer-efficiency result of (“Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026), which is consistent with, but does not resolve, the timing profile. (b) Optimal timing is plausibly equipment- and anthropometry-dependent (shaft flexibility, club MOI, segment lengths), and model sensitivity is untested across engines. (c) Whether *actively* restraining release (negative wrist torque early) outperforms merely *not driving* release is unresolved in 3D models with realistic actuators — Jorgensen’s and White’s analyses suggest passive holding suffices in 2D (T. P. Jorgensen 1999; White 2006), while Sprigings-Neal-type actuator models leave room for an active-restraint benefit (Sprigings and Neal 2000). These are precisely the questions the repository’s toolchain can answer.

## 9 Proposed experiments

The following designs are specified so that each has a pre-registered, falsifiable outcome measure. They are tracked in issue #8377; no results exist yet, and none are reported here.

**E1 — Torque-timing sweep (double pendulum and 8-DOF golfer).** Drive the models of Table 1 with parameterized torque programs: proximal torque fixed; wrist torque onset time  $t_{on}$ , sign profile (restrain-then-drive vs drive-only), and magnitude swept on a grid; impact defined by club alignment. Outcome: clubhead speed at impact as a function of  $t_{on}$  and profile. S2 predicts a late-interior optimum of  $t_{on}$  and a non-inferior or superior restrain-then-drive profile. The MJX/Warp batched backends make the sweep cheap; the ODE backend provides the reference.

**E2 — Counterfactual energy budget along swept trajectories.** For each E1 trajectory, evaluate the pointwise ZTCF/ZVCF and drift/control split (Equation 4) and integrate drift-side and control-side club power. Outcome: the fraction of final club kinetic energy attributable to drift vs control in early vs late downswing halves, as a function of performance. S2 predicts the drift share of late club energy gain rises with clubhead speed across the sweep.

**E3 — Re-simulation parity.** Implement the killswitch re-simulation ZTCF in Python on the backend-neutral model; compare against both the pointwise ZTCF profiles and the MATLAB pipeline on matched trajectories (cross-engine comparison service). Outcome: documented agreement/divergence between the two ZTCF operationalizations, closing the top gap of Table 2.

**E4 — Wrist-interface power accounting.** Implement Robertson-Winter segmental energy flow (joint force power  $\mathbf{F} \cdot \mathbf{v}$  plus moment power  $\tau \cdot \omega$  at each interface; issue #8378), then compute  $P_w(t)$  for E1’s best

and worst performers. Outcome: the direct time course of the transfer functional that Section 5 defines — the primary evidence either strategy needs.

**E5 — Mocap replication.** Run the E2/E4 analyses on measured swings ingested through the repository’s motion pipeline (C3D → tracked motion), across available skill levels. Outcome: observational test of the counterfactual predictions of Section 6 against human data, with the uncertainty-reporting conventions of the internal research review.

## 10 Limitations

This article synthesizes literature and specifies methodology; it reports no new data, and the S2 hypothesis, while strongly supported by the modeling literature, remains unquantified in the repository’s own terms until the experiments of Section 9 are executed. The kinematic-sequence evidence base carries known methodological sensitivity (Marsan et al. 2019); several classical results rest on planar models whose generalization to 3D swings with mobile hubs is a live modeling question (McPhee 2022); and the work-budget percentages cited from (Nesbit and Serrano 2005) derive from four subjects. One recent source (“Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026) is cited without its author list because the publisher’s site was unreachable through this environment’s network proxy at the time of writing; readers should resolve the DOI for full bibliographic details. Finally, energy-based arguments here concern clubhead *speed*; accuracy, consistency, and injury-risk trade-offs of late-transfer strategies (Hume, Keogh, and Reid 2005; Lindsay, Mantrop, and Vandervoort 2008) are outside scope.

## 11 Conclusion

Proximal-to-distal sequencing in the golf swing is real, robust, and — as a coaching target — chronically over-interpreted. The mechanical literature, from Cochran and Stobbs through modern optimal control, does not support “pass the energy along as soon as possible.” It supports a subtler program: *build the energy proximally with the club held folded, defer the handoff, and let the transfer run — mostly parametrically, with a brief late assist — when the centrifugal field can pay for it.* Skilled players differ from amateurs precisely along this axis: not in whether their segments peak in order, but in how much energy the proximal system holds at transition, how late and how efficiently it is handed to the club. The ZTCF/ZVCF counterfactual framework implemented in UpstreamDrift turns this qualitative account into measurable time profiles — drift versus control, passive versus active, early versus late — and the experiment suite specified here (epic #8373) is designed to either quantify the retain-early/release-late advantage or falsify it. Either outcome would be a contribution.

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## References

- Balzerson, Daniel, Joydeep Banerjee, and John MCPhee. 2016. “A Three-Dimensional Forward Dynamic Model of the Golf Swing Optimized for Ball Carry Distance.” *Sports Engineering* 19: 237–50. <https://doi.org/10.1007/s12283-016-0197-7>.
- Bunn, John W. 1972. *Scientific Principles of Coaching*. 2nd ed. Englewood Cliffs, NJ: Prentice-Hall.
- Cheetham, Phillip J., Gregory A. Rose, Richard N. Hinrichs, Robert J. Neal, Rob E. Mottram, Paul D. Hurrión, and Peter F. Vint. 2008. “Comparison of Kinematic Sequence Parameters Between Amateur and Professional Golfers.” In *Science and Golf v: Proceedings of the World Scientific Congress of Golf*,

- edited by Debbie Crews and Rafer Lutz. Mesa, AZ: Energy in Motion. <https://www.researchgate.net/publication/252236426>.
- Cochran, Alastair, and John Stobbs. 1968. *Search for the Perfect Swing*. Philadelphia: J. B. Lippincott.
- “Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing.” 2026. *Frontiers in Sports and Active Living* 8. <https://doi.org/10.3389/fspor.2026.1790645>.
- Han, Ki Hoon, Christopher Como, Jemin Kim, Sangwoo Lee, Jaewoong Kim, Dae Kyoo Kim, and Young-Hoo Kwon. 2019. “Effects of the Golfer–Ground Interaction on Clubhead Speed in Skilled Male Golfers.” *Sports Biomechanics* 18 (2): 115–34. <https://pubmed.ncbi.nlm.nih.gov/31042142/>.
- Hume, Patria A., Justin Keogh, and Duncan Reid. 2005. “The Role of Biomechanics in Maximising Distance and Accuracy of Golf Shots.” *Sports Medicine* 35 (5): 429–49. <https://doi.org/10.2165/00007256-200535050-00005>.
- Jorgensen, Theodore. 1970. “On the Dynamics of the Swing of a Golf Club.” *American Journal of Physics* 38 (5): 644–51. <https://doi.org/10.1119/1.1976419>.
- Jorgensen, Theodore P. 1999. *The Physics of Golf*. 2nd ed. New York: Springer / AIP Press.
- Lampsa, Michael A. 1975. “Maximizing Distance of the Golf Drive: An Optimal Control Study.” *Journal of Dynamic Systems, Measurement, and Control* 97 (4): 362–67. <https://asmedigitalcollection.asme.org/dynamicsystems/article-abstract/97/4/362/400950>.
- Lee, Junghoon, Justin Ehrlich, and Christopher Cain. 2026. “The Effects of Different Kinematic Sequences of the Elite Golf Swing on Clubhead Speed.” *International Journal of Sports Science & Coaching*. <https://doi.org/10.1177/17479541261429479>.
- Lindsay, David M., Shannon Mantrop, and Anthony A. Vandervoort. 2008. “A Review of Biomechanical Differences Between Golfers of Varied Skill Levels.” *International Journal of Sports Science & Coaching* 3 (Suppl. 1): 187–97. <https://doi.org/10.1260/174795408785024117>.
- MacKenzie, Sasho J., and Eric J. Springs. 2009. “A Three-Dimensional Forward Dynamics Model of the Golf Swing.” *Sports Engineering* 11 (4): 165–75. <https://doi.org/10.1007/s12283-009-0020-9>.
- Madrid, Braden, Marco A. Avalos, Nicholas A. Levine, Noelle J. Tuttle, Kevin A. Becker, and Young-Hoo Kwon. 2020. “Association Between the on-Plane Angular Motions of the Axle-Chain System and Clubhead Speed in Skilled Male Golfers.” *Applied Sciences* 10 (17): 5728. <https://doi.org/10.3390/app10175728>.
- Marsan, Thibault, Maxime Bourgain, Patricia Thoreux, Christophe Sauret, and Philippe Rouch. 2019. “Biomechanical Analysis of the Golf Swing: Methodological Effect of Angular Velocity Component on the Identification of the Kinematic Sequence.” *Acta of Bioengineering and Biomechanics* 21: 115–20. <https://www.researchgate.net/publication/333377321>.
- Marshall, Robert N., and Bruce C. Elliott. 2000. “Long-Axis Rotation: The Missing Link in Proximal-to-Distal Segmental Sequencing.” *Journal of Sports Sciences* 18 (4): 247–54. <https://doi.org/10.1080/026404100364983>.
- McNally, William, and John McPhee. 2018. “Dynamic Optimization of the Golf Swing Using a Six Degree-of-Freedom Biomechanical Model.” In *Proceedings*, 2:243. 6. <https://doi.org/10.3390/proceedings2060243>.
- McPhee, John. 2022. “A Review of Dynamic Models and Measurements in Golf.” *Sports Engineering* 25. <https://doi.org/10.1007/s12283-022-00387-0>.
- Meister, David W., Amy L. Ladd, Erin E. Butler, Betty Zhao, Andrew P. Rogers, Conrad J. Ray, and Jessica Rose. 2011. “Rotational Biomechanics of the Elite Golf Swing: Benchmarks for Amateurs.” *Journal of Applied Biomechanics* 27 (3): 242–51. <https://doi.org/10.1123/jab.27.3.242>.
- Milburn, Peter D. 1982. “Summation of Segmental Velocities in the Golf Swing.” *Medicine and Science in Sports and Exercise* 14 (1): 60–64. <https://pubmed.ncbi.nlm.nih.gov/7070260/>.
- Miura, Ken. 2001. “Parametric Acceleration — the Effect of Inward Pull of the Golf Club at Impact Stage.” *Sports Engineering* 4 (2): 75–86. <https://doi.org/10.1046/j.1460-2687.2001.00071.x>.
- Neal, Robert J., Ryan Lumsden, Mark Holland, and Bruce Mason. 2007. “Body Segment Sequencing and Timing in Golf.” *International Journal of Sports Science & Coaching* 2 (Suppl. 1). <https://doi.org/10.1260/174795407789705497>.
- Nesbit, Steven M. 2005. “A Three Dimensional Kinematic and Kinetic Study of the Golf Swing.” *Journal of Sports Science and Medicine* 4 (4): 499–519. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3899667/>.
- Nesbit, Steven M., and Ryan McGinnis. 2009. “Kinematic Analyses of the Golf Swing Hub Path and

- Its Role in Golfer/Club Kinetic Transfers.” *Journal of Sports Science and Medicine* 8 (2): 235–46. <https://www.jssm.org/volume08/iss2/cap/jssm-08-235.pdf>.
- Nesbit, Steven M., and Ryan S. McGinnis. 2014. “Kinetic Constrained Optimization of the Golf Swing Hub Path.” *Journal of Sports Science and Medicine*. <https://pubmed.ncbi.nlm.nih.gov/25435779/>.
- Nesbit, Steven M., and Monika Serrano. 2005. “Work and Power Analysis of the Golf Swing.” *Journal of Sports Science and Medicine* 4 (4): 520–33. <https://www.jssm.org/volume04/iss4/cap/jssm-04-520.pdf>.
- Pickering, W. M., and G. T. Vickers. 1999. “On the Double Pendulum Model of the Golf Swing.” *Sports Engineering* 2 (3): 161–72. <https://doi.org/10.1046/j.1460-2687.1999.00028.x>.
- Putnam, Carol A. 1991. “A Segment Interaction Analysis of Proximal-to-Distal Sequential Segment Motion Patterns.” *Medicine and Science in Sports and Exercise* 23 (1): 130–44. <https://pubmed.ncbi.nlm.nih.gov/1997807/>.
- . 1993. “Sequential Motions of Body Segments in Striking and Throwing Skills: Descriptions and Explanations.” *Journal of Biomechanics* 26 (Suppl. 1): 125–35. [https://doi.org/10.1016/0021-9290\(93\)90084-R](https://doi.org/10.1016/0021-9290(93)90084-R).
- Robertson, D. Gordon E., and David A. Winter. 1980. “Mechanical Energy Generation, Absorption and Transfer Amongst Segments During Walking.” *Journal of Biomechanics* 13 (10): 845–54. [https://doi.org/10.1016/0021-9290\(80\)90172-4](https://doi.org/10.1016/0021-9290(80)90172-4).
- Sharp, Robin S. 2009. “On the Mechanics of the Golf Swing.” *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 465 (2102): 551–70. <https://doi.org/10.1098/rspa.2008.0304>.
- Sprigings, Eric J., and Sasho J. MacKenzie. 2002. “Examining the Delayed Release in the Golf Swing Using Computer Simulation.” *Sports Engineering* 5 (1): 23–32. <http://sashomackenzie.com/publications/delayedrelease.pdf>.
- Sprigings, Eric J., and Robert J. Neal. 2000. “An Insight into the Importance of Wrist Torque in Driving the Golfball: A Simulation Study.” *Journal of Applied Biomechanics* 16 (4): 356–66. <https://doi.org/10.1123/jab.16.4.356>.
- Tinmark, Fredrik, John Hellström, Kjartan Halvorsen, and Alf Thorstensson. 2010. “Elite Golfers’ Kinematic Sequence in Full-Swing and Partial-Swing Shots.” *Sports Biomechanics* 9 (4): 236–44. <https://doi.org/10.1080/14763141.2010.535842>.
- White, Rod. 2006. “On the Efficiency of the Golf Swing.” *American Journal of Physics* 74 (12): 1088–94. <https://doi.org/10.1119/1.2346688>.
- Zajac, Felix E., and Michael E. Gordon. 1989. “Determining Muscle’s Force and Action in Multi-Articular Movement.” *Exercise and Sport Sciences Reviews* 17: 187–230. <https://pubmed.ncbi.nlm.nih.gov/2676547/>.
- Zajac, Felix E., Richard R. Neptune, and Steven A. Kautz. 2002. “Biomechanics and Muscle Coordination of Human Walking. Part I: Introduction to Concepts, Power Transfer, Dynamics and Simulations.” *Gait & Posture* 16 (3): 215–32. [https://doi.org/10.1016/S0966-6362\(02\)00068-1](https://doi.org/10.1016/S0966-6362(02)00068-1).
- Zheng, Naiquan, Steve W. Barrentine, Glenn S. Fleisig, and James R. Andrews. 2008. “Kinematic Analysis of Swing in Pro and Amateur Golfers.” *International Journal of Sports Medicine* 29 (6): 487–93. <https://doi.org/10.1055/s-2008-1038732>.